Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 5.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. Simplify:

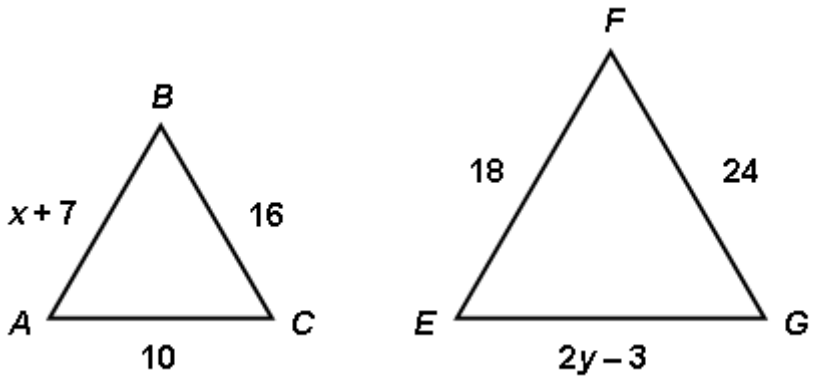
$$3x\left(3x^{-\frac{1}{3}}\right)^3$$

A) $\frac{1}{9}$ B) 1 C)81 D) $\frac{x^2}{9}$ E) x^2 F) $81x^2$

2. The first term of the arithmetic of the arithmetic sequence is 8 and $16th$ term is -37. Find a common difference.

A) 3 B) -3 C) 2 D) -2 E) -4

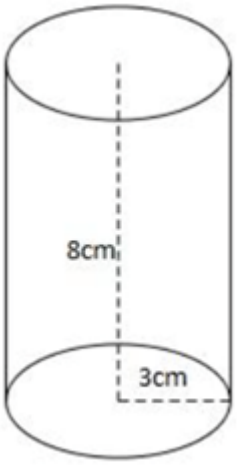
3. In the figure shown, ΔABC and ΔEFG are similar triangles.



Find $x + y$.

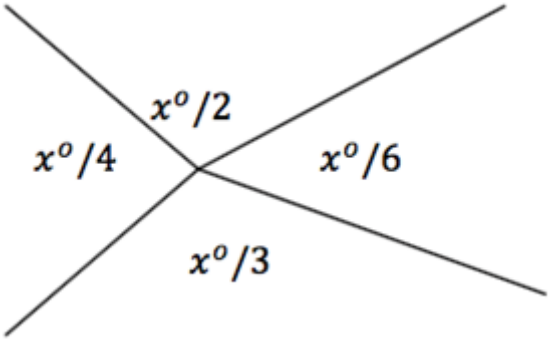
A) 10 B) 12 C) 14 D) 16 E) 18

4. Cylinder in a picture below has a radius of $3\ cm$ and height of $3\ cm$. If cylinder is closed from both ends, find its surface area.



A) $72\pi cm^2$ B) $60\pi cm^2$ C) $66\pi cm^2$ D) $24\pi cm^2$ E) $82\pi cm^2$

5. What is the value of x?



A) 180 B) 288 C) 336 D) 240 E) 312

6. Assume that every 6-digit number is positive telephone number except those that begin with 0 or 1. What fraction of telephone numbers begin with 3 and end with 5?

A) $1/36$ B) $1/60$ C) $1/80$ D) $1/90$ E) $1/100$

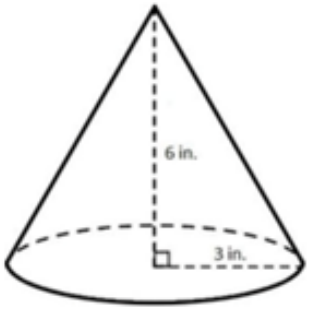
7. A lighthouse emits a light which can be seen for 60 miles in all directions. If the intensity of the light is strengthened so that its visibility is increased by 40 miles in all directions, by approximately how many square miles is its region of visibility increased?

A) $6{,}300$ B) $10{,}000$ C) $10{,}300$ D) $20{,}000$ E) $31{,}400$

8. When the positive integer n is divided by 9, the remainder is 5. When $n + 29$ is divided by 9, what is the remainder?

A) 7 B) 2 C) 5 D) 6 E) 4

9. Cone shown below has a radius of 3 inches and height of 6 inches. Find the volume of the cone in cubic centimeters. $1\ inch\ =\ 2.5\ cm$



A) $45\pi\ cm^3$ B) $281.25\pi\ cm^3$ C) $290.25\pi\ cm^3$ D) $60.75\pi\ cm^3$ E) $72.4\pi\ cm^3$

10. Find the sum of the first 6 prime numbers

A) 6 B) 24 C) 34 D) 41 E) 52

11. Simplify

$$\left(\frac{2xy^{(-2)}y^4}{4xy}\right)^{(-1)}\left(\frac{xy}{2x^3y}\right)$$

A) $4/(xy)$ B) x^2y C) $\frac{1}{x^2*y}$ D) $\frac{1}{xy^2}$ E) x^2y^2

12. A triangle is to be changed by increasing the length of its base by 40% and decreasing the length of its height by 40%. Which of the following must be true about the area of the triangle?

A) It will increase by 40% B) It will increase by 16% C) It will not change D) It will decrease by 16% E) It will decrease by 40%

13. P, Q and R are each mixtures of red and white paint. The percentage by volume of red paint in P is 30%. The percentage by volume of red paint in Q is 20%. The mixtures P, Q, and R are combined in the proportion $12:5:3$ are respectively.

If the resulting mixture contains 25% by volume of red paint, what percentage by volume of mixture R is red paint?

A) 25% B) 23% C) $13\frac{1}{3}\%$
D) $19\frac{1}{2}\%$ E) $9\frac{3}{4}\%$ F) It is impossible to achieve this result.

14. The line segment joining the points $(3,3)$ and $(7,5)$ is a diameter of a circle.

This circle is translated by 3 units in the negative - x direction, then reflected in the x–axis, and then enlarged by a scale factor of 4 about the center of the resulting circle.

The equation of the final circle is

A) $(x-2)^2+(y-4)^2=320$ B) $(x-2)^2+(y+4)^2=320$ C) $(x-2)^2+(y-4)^2=80$
D) $(x-2)^2+(y+4)^2=80$ E) $(x-2)^2+(y-4)^2=20$ F) $(x-2)^2+(y+4)^2=20$

15. $11+22+33+44+55=759$

In order to obtain a correct equation it is permitted to multiply any 5 numbers on the left hand side of the equation by 10. For example, $110+22+33+44+550=759$. How many different ways of making the equation above correct are there?

A) 1 B) 2 C) 3 D) 4 E) 5

16. The sequence $a_n = (-1)^n - (-1)^{n-1} + (-1)^{n+2}$ for $n \geq 1$.

Find the value of

$$\sum_{n=1}^{39} a_n$$

A) -39 B) -3 C) -1 D) 0 E) 1 F) 3 G) 39

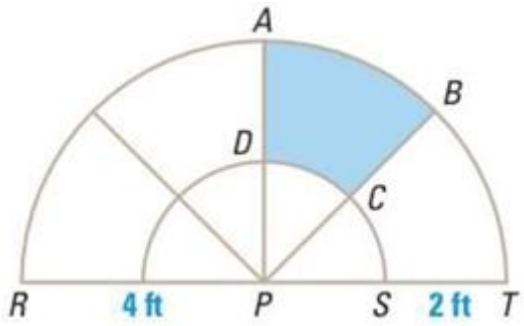
17. In a sequence the first term is 1 and each term is twice the previous term plus one. Find the sum of the first five terms of the given sequence.

A) 31 B) 57 C) 62 D) 49 E) 38

18. Find the ratio of the circumference of a circle with radius r to its area.

A) $r/2$ B) π/r C) $\frac{2}{\pi}$ D) $2/r$ E) $\frac{\pi}{2}$

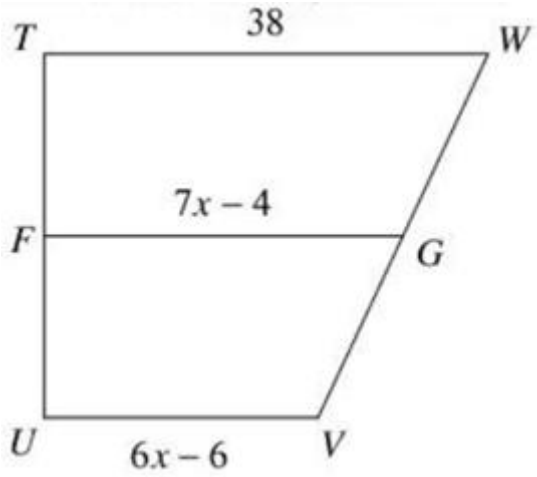
19. Find the area of the shaded region. $PR = 4\ ft$, $ST = 2\ ft$, $\angle APB = \pi/4$



Choose one

A) $1.5\pi\ ft^2$ B) $2.0\pi\ ft^2$ C) $2.5\pi\ ft^2$ D) $3.0\pi\ ft^2$ E) $3.5\pi\ ft^2$

20. FG is the median of the trapezoid. Find UV.



A) 5 B) 24 C) 20 D) 31 E) 38

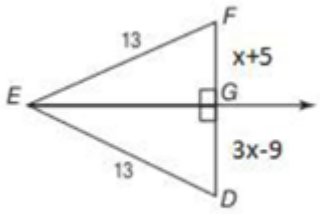
21. Let $a\#b\#c$ be defined by equation $a\#b\#c = \left(\frac{a}{c}\right) \times b + a \times c$ for all non-zero numbers a, b and c. Which of the following is equal to an odd integer?

A) $4\#8\#2$ B) $3\#2\#1$ C) $9\#3\#3$ D) $8\#6\#4$ E) $5\#7\#1$

22. At a fancy vegetable stand, Tom bought purple tomatoes for $\$8$ each and Jerry bought organic giant cabbages for $\$12$ each. In total, they played $\$104$ for 10 vegetables. How many cabbages did Jerry buy?

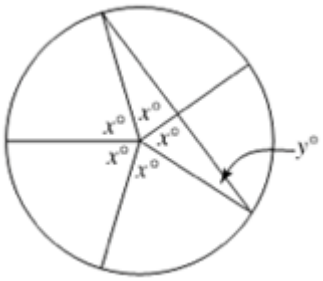
A) 4 B) 5 C) 6 D) 7 E) 3

23. Find EG



A) 12 B) 5 C) 7 D) 8 E) 10

24. In the figure below, what is the ratio of y to x?



A) $\frac{1}{5}$ B) $\frac{1}{4}$ C) $\frac{1}{3}$ D) $\frac{1}{2}$ E) It cannot be determined from the information given.

25. For any positive integer $n > 1$, $n!$ Represents the product of the first n positive integers. For example, $3! = 1 \times 2 \times 3 = 6$. Which of the following is (are) equal to $\frac{10!}{8!}$?

I. $5! - 4! - 3!$

II. $\frac{5!}{4!}$

III. $15(3!)$

A) I only B) II only C) III only D) I and III only E) I, I and III.

26. Which of the following best describes the graph of the function $f(x) = 5x - 7$?

A) A horizontal line

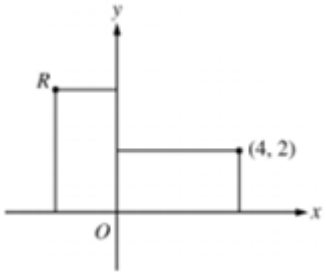
B) A line that increases from left to right

C) A line that decreases from left to right

D) A parabola that opens up

E) A parabola that opens down

27. If the areas of the rectangles in the figure below are equal, which of the following could be the coordinates of point R?



A) $(-2, -4)$ B) $(-2, -2)$ C) $(-2, 4)$ D) $(2, -4)$ E) $(2, 4)$

28. If $\frac{1}{6}+\frac{1}{7}+\frac{1}{8}>\frac{1}{x}+\frac{1}{7}+\frac{1}{8}$, then x could be which of the following?

A) 3 B) 4 C) 5 D) 6 E) 7

29. $2\#5 =$

Let # be defined by $a\#b = ab + a + b$ for all numbers a and b.

A) 7 B) 10 C) 17 D) 25 E) 27

30. In the equation below, k is a constant. If the roots of the equation are 7 and 2, what is the value of k?

$$3(x-7)(x-2) = k$$

A) 0 B) 2 C)3 D) 7 E) 14

Answer key:

№	Correct Answer	Your answer
1	C	
2	B	
3	C	
4	C	
5	B	
6	C	
7	D	
8	A	
9	B	
10	D	
11	C	
12	D	
13	C	
14	B	
15	B	
16	B	
17	B	
18	D	
19	A	
20	B	
21	B	
22	C	
23	B	
24	B	
25	D	
26	B	
27	C	
28	E	
29	C	
30	A	
		Total: _____